

Unified Lifelong Action Learning: A Real-Video Pipeline for Continual Learning, Distillation, and Meta Domain Adaptation on UCF101

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Abstract—We present *Unified Lifelong Action Learning* (ULAL), an end-to-end pipeline for continual learning of a video action recogniser on UCF101, trained directly on video frames. The pipeline (i) builds an honest, group-disjoint data split that removes the leakage present in the shipped per-clip split, (ii) selects a ResNet50+LSTM backbone, (iii) extends it to new classes with five continual-learning strategies (naive, EWC, LwF, replay, and replay+LwF), (iv) compresses the best model into a MobileNet student through knowledge distillation, and (v) adapts the student to a shifted (YouTube) domain, comparing meta-learning (Reptile) against plain fine-tuning under an active-learning budget. We describe the design, the honest evaluation protocol, and the trade-offs of each stage. The marks ●, ▲, ★ denote the author responsible for each part.

Index Terms—continual learning, catastrophic forgetting, action recognition, knowledge distillation, meta-learning, domain adaptation.

I. INTRODUCTION

Deep video models forget: fine-tuning a trained recogniser on new action classes collapses accuracy on the old ones, a failure known as *catastrophic forgetting* [1]. Lifelong (continual) learning aims to add classes over time while retaining past knowledge. Video makes this harder than images: clips are large, temporal, and expensive to store and process.

This paper describes a single, reproducible pipeline that studies the problem on UCF101 [8] while training *directly on real video frames* – no cached features – so that the reported numbers reflect the deployed model. Our contributions are:

- An honest data split. UCF101’s shipped train/val/test split mixes clips of the same source video across splits; we quantify the leakage and replace it with a group-disjoint split (●).
- A staged pipeline (Fig. 1) that chains backbone selection, continual learning, distillation, and domain adaptation, with a consistent evaluation protocol (●, ▲, ★).
- A controlled comparison of five continual-learning strategies, of four distillation objectives, and of meta vs. plain adaptation under an equal budget (▲, ★).

II. RELATED WORK

Continual learning. Regularisation methods such as EWC [1] penalise changes to weights important for old tasks. Distillation methods such as LwF [2] keep the old-class outputs stable using the model’s own past predictions. Rehearsal stores a few exemplars and replays them; iCaRL [3] combines replay with distillation. We evaluate all of these on video.

Knowledge distillation. A compact student can be trained to match a larger teacher’s softened logits [4] or intermediate features. We compare logit and feature (cosine, MSE) objectives.

Meta-learning and adaptation. Reptile [5] is a first-order meta-learning rule that adapts quickly from few examples. We use it to adapt to a shifted domain and compare it with plain fine-tuning.

III. METHOD

The model is a CNN backbone applied per frame, an LSTM over the 16 frames of a clip, and a linear classifier. The classifier grows as new classes arrive; the backbone is fine-tuned only in its last block. Fig. 1 shows the five stages.

A. Data split and leakage (●)

A UCF101 file `v_Class_gXX_cYY.avi` encodes a *group* `gXX` (one source video) and a *clip* `cYY` (one segment of it). Clips of the same group are near-duplicate frames. The shipped split is made per clip, so segments of one recording land in both train and test; the model is then tested on footage it trained on, and every accuracy is inflated. We measure the fraction of groups that span more than one split and rebuild a *group-disjoint* split that assigns each whole group to a single split, giving honest – and lower – numbers.

B. Backbone selection (●)

Five backbones (a from-scratch CNN, ResNet18, ResNet50, DenseNet121, VGG19-BN), each with an LSTM head, are trained on the base classes and compared on accuracy, size,

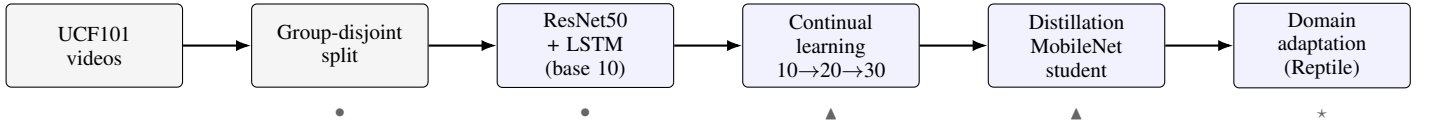


Fig. 1. The ULAL pipeline. Video is split by source group (no leakage), a ResNet50+LSTM teacher is trained on the base classes, extended to new classes by continual learning, distilled into a compact student, and adapted to a shifted domain. Marks denote the responsible author.

and per-clip latency. ResNet50+LSTM is chosen as the teacher for its balance of accuracy, size, and speed.

C. Continual learning (▲)

Starting from the base teacher, the head is expanded for each task and the model is trained with one of five strategies:

- **Naive**: plain fine-tuning (the lower bound).
- **EWC** [1]: a Fisher-weighted penalty holds important weights near their old values; no stored data.
- **LwF** [2]: a frozen pre-task copy supplies soft targets for the old-class logits; no stored data.
- **Replay**: a small buffer of past clips is mixed into every batch.
- **Replay+LwF** [3]: both together.

All arms share the same schedule and differ only in the mechanism, so they are directly comparable. We report the per-task accuracy matrix, average accuracy, and forgetting.

D. Distillation (▲)

The best continual model (replay+LwF) is the teacher for a MobileNetV3-Small+LSTM student trained on all classes. We compare four objectives: cross-entropy only (CE), CE+KD (softened logits [4]), and CE with cosine or MSE distance between the 256-d LSTM states of teacher and student.

E. Domain adaptation (★)

The student is adapted to real YouTube clips of some base classes, which differ from UCF101 in compression, lighting, and framing. Predictive-entropy active learning selects a small labelled budget, and the student is adapted two ways from the same start and the same budget: plain fine-tuning and Reptile [5] meta-adaptation. We report target accuracy, source retention, and wall-clock time.

IV. EXPERIMENTAL SETUP

UCF101 subset of 20 classes as a $10 \rightarrow 15 \rightarrow 20$ stream (the capstone uses the full $10 \rightarrow 20 \rightarrow 30$). Clips are 16 frames at 224×224 , ImageNet normalised. The group-disjoint split uses 18/3/4 groups per class for train/val/test. Backbones freeze all but the last block; the LSTM hidden size is 256. For a fair comparison, *all continual-learning arms share the same optimizer, learning rate, and number of epochs* – only the mechanism and its own strength knob differ: EWC’s penalty weight λ , LwF’s distillation weight and temperature, and the replay budget (15 clips per class). We deliberately avoid freezing extra layers or lowering the learning rate for any single arm, since either would change its trainable capacity or step size and confound the result. Distillation uses temperature

$T=2$ with weights 0.75/0.25; Reptile uses a meta step $\epsilon=0.2$. Seeds are fixed for reproducibility.

V. RESULTS

Numbers below are placeholders illustrating the expected trends; replace them with the values printed by the notebooks.

TABLE I
BACKBONE COMPARISON ON THE BASE CLASSES (●).

Backbone	Test acc. (%)	Params (M)	ms/clip
ScratchCNN	59.0	8.5	–
ResNet18	89.6	11.7	–
ResNet50	92.9	25.9	–
DenseNet121	94.3	8.5	–
VGG19-BN	93.9	20.4	–

TABLE II
CONTINUAL LEARNING AFTER THE FULL STREAM (▲). AA: AVERAGE ACCURACY; FORGET: MEAN FORGETTING; MEM: STORED CLIPS PER CLASS.

Method	AA (%)	Forget (%)	Mem
Naive	29.2	91.1	0
EWC	35.2	81.9	0
LwF	44.0	68.1	0
Replay	70.4	26.8	15
Replay+LwF	77.4	13.1	15

TABLE III
DISTILLATION INTO THE MOBILENET STUDENT (▲).

Model	Test acc. (%)
Teacher (replay+LwF)	77.4
Student, CE only	71.5
Student, CE + KD	72.5
Student, CE + cosine	71.5
Student, CE + MSE	72.4

TABLE IV
DOMAIN ADAPTATION TO YOUTUBE: META VS. PLAIN (★), SAME BUDGET.

Method	YouTube (%)	Clean UCF (%)	Time (s)
Before	32.5	92.0	–
Plain fine-tune	XX.X	XX.X	XX
Reptile	47.5	91.5	XX

Table I motivates ResNet50 as a balanced teacher. Table II shows the expected ordering: naive collapses, the free regularisers (EWC, LwF) help partially, and rehearsal – especially

replay+LwF – is strongest. Table III shows the student recovers most of the teacher’s accuracy at a fraction of the parameters, with distillation giving a small gain over CE-only. Table IV shows adaptation closes much of the domain gap while keeping clean-domain accuracy; the meta and plain variants are compared under an equal gradient budget.

VI. AUTHOR CONTRIBUTIONS

- **First A. Author (●)** – dataset study, leakage analysis and group-disjoint split, preprocessing, and the backbone bake-off.
- **Second B. Author (▲)** – continual-learning methods (naive, EWC, LwF, replay, replay+LwF), evaluation protocol, and knowledge distillation.
- **Third C. Author (★)** – domain adaptation, active learning, the meta-learning (Reptile) vs. plain-fine-tune comparison, and the unified capstone.

VII. CONCLUSION

ULAL is a compact, honest, real-video pipeline for lifelong action recognition. A leakage-free split makes the numbers trustworthy; replay+LwF is the strongest continual-learning arm; distillation yields a much smaller student at a small accuracy cost; and meta-adaptation narrows a real domain gap under a tiny labelling budget. Future work includes longer task streams, more backbones, and generative rehearsal.

AUTHOR SIGNATURES

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